4	**(a)**	**(i)**	$R = \frac{\rho l}{A} = \frac{1.1\times10^{-6}\times98}{\pi(\frac{1.1\times10^{-3}}{2})^2}$ = 113 Ω		A1
		(ii)	As the wire stretches, its length increases and it cross sectional area decreases. From $R = \frac{\rho l}{A}$, R will increase.		B1
	(b)	**(i)**	$R_{PQ} = 1\% \times 113 = 1.13\Omega$ $V_{PQ} = \frac{R_{PQ}}{R_{PQ}+R+r} \times E_1$ $V_{PQ} = \frac{1.13}{1.13+R+r} \times E_1$ V_{PQ} per unit length $= \frac{\frac{1.13}{1.13+R+r}\times E_1}{0.98}$ $= \frac{1.15}{1.13+R+r} \times E_1$		M1 C1 for correct R and length of PQ A1
			1.	Length PJ = 1/3 x 0.98 = 0.327 m Using (b)(i) ans, $V_{PJ} = 0.327\times\frac{1.15}{1.13+R+r}\times E_1$ $= 0.327\times\frac{1.15}{1.13+1.0+r}\times 12$ At balanced length, $V_{PJ} = E_2$ $0.327\times\frac{1.15}{1.13+1.0+r}\times 12 = 1.5$ r = 0.878 Ω	C1 C1 A1
			2.	If the resistance R is too high, the pd across the PQ will be too low. If the p.d. across PQ is lower than the e.m.f. of E_2, there will be no balance point.	B1 B1